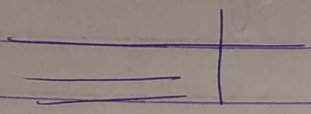
Logic fun^n:

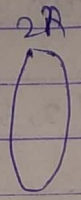
→ $xy + \bar{x}z + yz = xy + \bar{x}z$ → Consensus Theorem

→ principle of Duality : $+ \to \cdot$ & $0 \to 1$
$\cdot \to +$ $1 \to 0$

→ n-boolean variable



2^n entries



⇒ # fun^n : 2^{2^n}

∵

1	2	3	—	n
k	k	k		k
		k^n		

n values

m^{k^n} ⇒ # fun^n

→ Neutral fun^n:

Minterms = # Maxterms

$^{2^n}C_{(2^n/2)}$.

total 2^n.

select $2^n/2$. ++

Eg:

Using NAND:
EX-OR : 4
EX-NOR : 5

Using NOR:
EX-OR : 5
EX-NOR : 4

→ functionally Complete:

$(+, \cdot, -)$ All take support
$(+, -)$ from 0 & 1, then
$(\cdot, -)$ Partially FC.

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Self dual: $f(ABC) = AB + BC + CA$

dual → $f_d(ABC) = (A+B)(B+C)(C+A)$

if $f = f_d$ → then self dual.

→ fun^n should be neutral. [i.e. # minterms = # maxterm].

→ fun^n should not contain two mutually exclusive terms.

eg. $(0,7)(1,6)(2,5)(3,4)$

Any one. ✓

eg. $AB + BC + CA$

↳ Canonical form: $ABC + AB\bar{C} + A\bar{B}C + \bar{A}BC$

$AB\bar{C}$ should not present. check for All → Mutual Exclusive

self dual fun^n: $2^{2^{n-1}}$ → # minterm = # maxterm

2^{n-1} ⇒ $2^{2^{n-1}}$ ++

Ques. $f(w,x,y,z) = wx + yz$.

↳ Convert in Canonical form = 9 minterms.

So, # fun^n covering f.

2^n → total minterms

2^{16} → $16 - 7 = 9$.

fun^n $= 2^9$. ++

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Que. Add to make self dual-

$f(A,B,C,D) = \bar{A}BC + (AC+B)D$

$\bar{A}BC(D+\bar{D}) + A(B+\bar{B})C(D+\bar{D}) + (A+\bar{A})B(C+\bar{C})D.$

7, 6, 15, 11, 5, 13

(0, 15) (1, 14) (2, 13) (3, 12) (4, 11) (5, 10) (6, 9) (7, 8)

2 2 1 1 4 Aus.

~~2 x 2 x 2 = 8~~

(1, 3) or (1, 12) or (14, 12) or (14, 3).

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Minimization:

→ An irredundant expression not necessary minimal always & nor the minimal expression is unique.

→ Prime Implicants [Not covered by any other cube]

→ Essential prime implicants. [doesn't have any other alternate to include.]

++

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Combinational Circuit

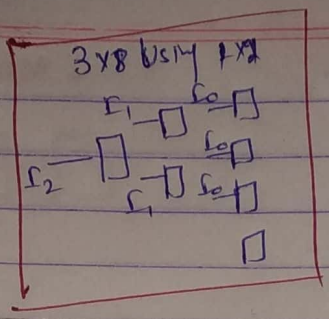
MUX. :

(4×1) for 32×1

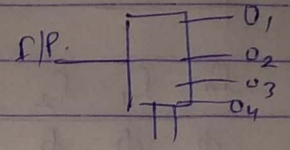
$\lceil 32/4 \rceil = 8$

$\lceil 8/4 \rceil = 2$

$\lceil 2/4 \rceil = 1$



DE-MUX :

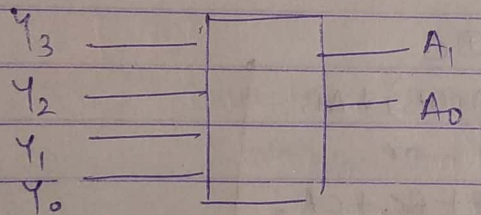


S_1 S_2

S_1	S_0	O_4	O_3	O_2	O_1
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

devices $\sum_{k=1}^{\lceil \log_2 M \rceil} (M/N^k)$ & # levels for M×1 by N×1 $\lceil \log_N M \rceil$

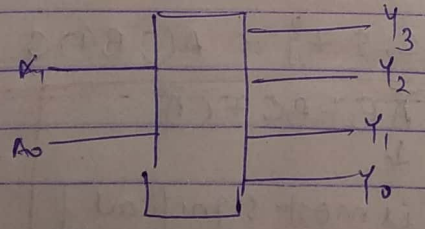
En-Coder: $2^n \times n$, used in lossless Compression.



Y_3	Y_2	Y_1	Y_0	A_1	A_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

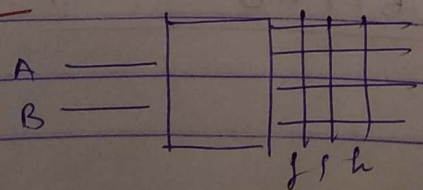
Decoder: $n \times 2^n$

↳ Also check for enable bit. If Enable is zero. Then O/P 'Dio.



A_1	A_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

ROM :



Comm^n = Horizontal line + Vertical line

Intersection = HL * VL.

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Priority Encoder: $I_3 > I_2 > I_1 > I_0$

I_0	I_1	I_2	I_3	O_1	O_0
ϕ	ϕ	ϕ	1	1	1
ϕ	ϕ	1	0	1	0
ϕ	1	0	0	0	1
1	0	0	0	0	0

Adder:

Half-Adder:

Sum: $A \oplus B$

Carry: AB

Full Adder:

Sum: $A \oplus B \oplus C$

Carry: $C(A \oplus B) + AB$

or

$AB + BC + CA$.

Subtractor:

Half-sub:

Diff: $A \oplus B$

Borrow: $\bar{A}B$

full sub:

Diff: $\{1, 2, 5, 7\} = A \oplus B \oplus C$

Borrow: $\bar{A}B + BC + C\bar{A}$

↓

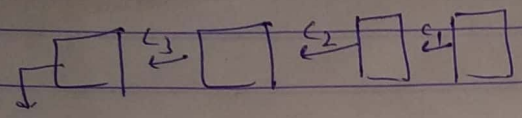
[A is most significant]

or

$C(A \oplus B) + \bar{A}B$

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Ripple Carry Adder:



Carry look-Ahead:

$$C_1 = C_0(A_0 \oplus B_0) + A_0B_0$$

So,

$$A_i \oplus B_i = P_i$$
$$A_i B_i = G_i$$

** Imp.

⇒ the no of AND Gates in carry generator circuit is $= \frac{n(n+1)}{2}$ [of Any i/p]

⇒ # OR GATE $= n$.

Imp Ques:

Consider a carry lookahead adder for adding two n-bit integers, but using gates of fan-in atmost two. The time to perform addition is-

$O(\log n)$

fan in = # inputs then $O(1)$

Que. 2.24 Previous Year **

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Sequential Circuit

→ positive Level → Negive Level
→ +ive edge → -ive edge
(→ Edge) *

• SR-FF:

$Q_{n+1} = S + \bar{R}Q_n$

fun" Table

S	R	Q_{n+1}
0	0	Q_n [Latch]
0	1	0
1	0	1
1	1	Invalid

Exitation table

Q_n	Q_{n+1}	S	R
0	0	0	ϕ
0	1	1	0
1	0	0	1
1	1	ϕ	0

• JK-FF

$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$

fun" table

J	K	Q_{n+1}
0	0	Q_n [Latch]
0	1	0 [Reset]
1	0	1 [Set]
1	1	$\bar{Q}_n$ [toggle]

Exitation table

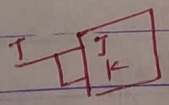
Q_n	Q_{n+1}	J	K
0	0	0	ϕ
0	1	1	ϕ
1	0	ϕ	1
1	1	ϕ	0

• T-FF

$Q_{n+1} = \bar{T}Q_n + T\bar{Q}_n$

fun" table

T	Q_{n+1}
0	Q_n
1	$\bar{Q}_n$



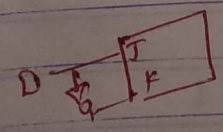
Excitatin table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

• D-FF

fun" table

D	Q_{n+1}
0	0
1	1

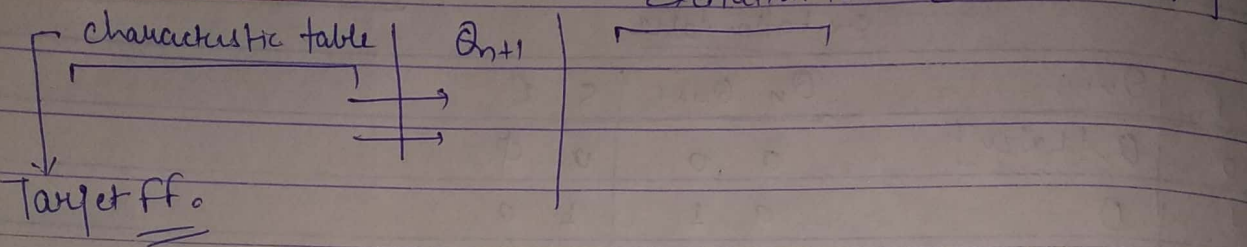


Excitation table

Q_n	Q_{n+1}	D
0	0	0
0	1	→ Not possible
1	0	→ Not possible
1	1	1

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Conversion b/w ff:



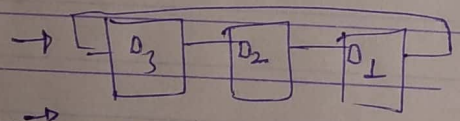
Counters:

→ Both are part of shift Counter.

Ring:

→ Using 'N' ff - Mod N.

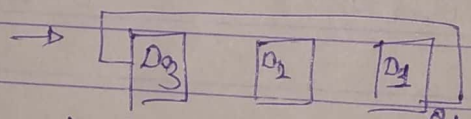
→ $I_0 = Q_{N-1}$



Johnson

→ Using 'N' ff - Mod 2N.

→ $I_0 = \overline{Q}_{N-1}$



Least Gets to Most

→ 0 → 1 → 3 → 7 → 5 → 4 At

✓ Self starting Counter: if possible to enter Counting loop irrespective of the initial state.

✓ Free Running Counter: if it contains all possible states in Counting loop.

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-ive Edge : up Counter
+ive Edge : down Counter.

→ n-bit Ripple Counter can Count upto 2^n. (**)

Ripple / Asyn Counter :

↳ frequency : $\frac{\text{forginal}}{2^{\#ff.}}$

Eg.

$Q_{0N} = \overline{Q_0}$: Every clock
$Q_{1N} = \overline{Q_1}$; when Q_0 [1 → 0]
$Q_{2N} = \overline{Q_2}$; when Q_1 [1 → 0].

→ only in these Cond^n we have to change the o/p, otherwise Latch.

↳ Mod 8 up : 0 → 1 → 2 → - - 7
↳ Mod 8 down : 7 → 6 → 8 - - 1 → 0.

Truncate Ripple Counter : if a MOD of Less than 2^n is req. eg.

3, 2, 0, 3, 2, 0 - etc.

Imp Ques : # J-K ff Req. for 0-1-0-2-0-3 states?

(**) total distinct states = 4 [0, 1, 2, 3]

So, # Bits = 2
So, # ff = 2
And to distinguish 3 zero's [0, 0, 0]
⇓ 2 More ff.

Ans. So, 4 ff

(**) OR ff = 3 (**) $2^K \geq n$. ⇒ [n = 6]
So, K = 3.

Ques : The Min^m No of J-K ff Req. to Construct a syn counter (0, 0, 1, 1, 2, 2, 3, 3, 0, 0 - -) ??

0, 1, 2, 3 → 2 ff.
00, 11 → to distinguish
0, 0 → 1 ff.
So, total 3 ff (**)

000 → 100 → 001 → 101 → 010 → 110 → 011 → 111
All distinguish.

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Que # FF required to design mod-258 Counter-

$\lceil \log_2[258) \rceil = 9$ ff. [keep Accuracy].

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Number System

Que What is the min^{u} no of digits required to represent a 32-digit in decimal no in binary.

[Base b]

k-digits & Base b.

$b^k - 1 \leq x^n - 1$

$n \geq k \log_x b$

$n \geq 32 \log_2 10$ ✓

or

decimal. Binary

$10^{32} - 1 \leq 2^n - 1$

Que $(123)_x = (122)_3$

$x > 3$ ↓ $x < 3$

Contradict / Not possible.

Que Roots of $x^2 - 11x + 22 = 0$ are 3 & 6 then base?

$x^2 - (b+1)x + (2b+2) = 0$

put $x = 3$

$b = 8$

same for $x = 6$.

Que -3 to 3 as [C, B, A, 0, 1, 2, 3] then $(102) = ?$

No = Base (7).

7	102		
7	15	-3	C
7	2	1	1
	0	2	2

(C12) Ans.

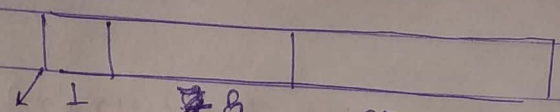
सरल भाषा मे लिखित

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Floating Point Representation:

↳ Explicit: $(-1)^S \times 0.M \times 2^{E\text{-Bias}}$ ↳ Implicit: $(-1)^S \times 1.M \times 2^{E\text{-Bias}}$

32-bit.



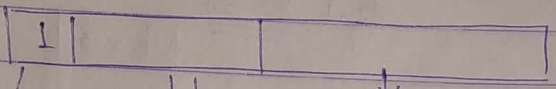
1 | 8 | 23

Sign | Exponent | fraction

⇒ Single precision.

Bias 127. ⟶ $(2^{n-1}-1)$ ++

64-bit



1 | 11 | 52 bits

⇒ Double precision.

Sign bit. | $1(2^{10}-1)$ | 52 bits

Bias = 1023

Imp: Single Precision don't have any representation / expression for '0'.

Sign	E	M	Value
0/1	0	0	±0
	127+128 (255)	0	±∞
0/1	E≠0 or 1≤E≤254	M≠0	Implicit $(-1)^S \times (1.M) \times 2^{E\text{-Bias}}$
0/1	E=0	M≠0	fraction
	E≠0 bcoz E=255	M≠0	NAN.

Same Applicable for double precision with diff limits.

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Compliment operation

↳ subtraction can be done using Addition with Compliment.

Base b → (b-1) → $(b^n-1)-x$

→ (b) → $(b^n-1)-x+1 = b^n-x$.

Also called Radix Compliment.

diminished Radix.

Significant of (b-1)'s compliment: It gives negation of x.

(b-1)'s of $x = \bar{x}$.

Sup. $(x-y)$

$[x+(-y)] \Rightarrow x+[(b^n-1)-y]$

$\Rightarrow (b^n-1)+(x-y)$

It should be zero.

(b-1)'s Compliment (1's Compliment)

→ Zero can be represent in two ways. [All zero, All 1's].

Sup

eg. base = 3.

$3^n-1 = 222 \ldots$ n bits = 0

$000 \ldots$ n bits = 0.

→ subtraction: Video 14.

$(x-y)$

$\Rightarrow (x+\bar{y})$

$\Rightarrow [x+(10^2-1-y)]$

$\Rightarrow [x+y]\ 99+(x-y)$

↓

If carry generate then

$99+(x-y)-100$

$(x-y)-1+1$

b's Compliment (Radix Compliment)

→ $(x-y)$

$[x+(-y)] \Rightarrow x+[b^n-y]$

$\Rightarrow b^n+(x-y)$

It should be zero.

→ So, only single pattern to represent zero.

Sup

→ subtraction:

eg. base = 10

97 − 23 = ?

? $97-23 = 97+(\bar{23})$

$= 97+(99-23+1)$

$= 97+77$

$= 174$

↓

Carry generate.

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सरल भाषा मे लिखित
का स्पष्टीकरण सहित सॉल्वड पेपर

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$(x+\bar{y}) = (b^n-1)+(x-y)$

$(x-y) \geq 1 \Rightarrow$ +ive → Carry generate

$(x-y) \leq 0 \Rightarrow$ No carry.

eg. $97-23 = 97+(\bar{23})$

$99-23 = 76$

$\Rightarrow 97+76 = 173$

Carry generate, so Remove & Add.

$\begin{array}{r} 173 \\ +\quad 1 \\ \hline 174 \end{array}$ $\quad$ $\begin{array}{r} 73 \\ +\quad 1 \\ \hline 74 \end{array}$

At

→ if carray generate, then Add it to Result.

→ if not get any carry, then do again compliment. That will be -ive of Result.

eg. $23-97 \Rightarrow 23+(\bar{97}) = 23+(2)$

$= 25$

No Carry, so $99-25 = -74$ Ans.

✓ -ive No. [Additive inverse]

$-x \Rightarrow (b^n-1)-x$

$(x+\bar{y}) = b^n+(x-y)$

$(x-y) \geq 0$ Carry

$(x-y) < 0$ No Carry. ①

Ex.

→ if Carry generate, then ignore that.

→ if no carry generate, then find Again b's Compliment.

✓ -ive No. [Additive Inverse]

$\Rightarrow -x \Rightarrow (b^n)-x$.

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Number System Continue...

- Binary
  - Unsigned
    - Weighted
    - 0 to $2^n - 1$
  - Signed Number
    - Sign Magnitude Form
      - Weighted
      - $(-2^{n-1}+1$ to $2^{n-1}-1)$
      - diff H/w Req for Add & Sub.
    - Sign 1's Complement
      - Not weighted
      - $|-(2^{n-1}-1)$ to $(2^{n-1}-1)|$
      - $x - y = x + \bar{y}$ ⇒ -ve No = 1's Complimet.
      - Same H/w used for Add & Sub.
    - Signed 2's Complement.
      - Weighted.
      - $|-(2^{n-1})$ to $(2^{n-1}-1)|$

$|-2^n \; \cdots \; 2^2 \; 2^1 \; 2^0|$ Ans.

	Unsigned	Sign	1's	2's
0 0 0	0	0	0	0
0 0 1	1	1	1	1
0 1 0	2	2	2	2
0 1 1	3	3	3	3
1 0 0	4	-0	-3	-4
1 0 1	5	-1	-2	-3
1 1 0	6	-2	-1	-2
1 1 1	7	-3	-0	-1

का समावेश ◉ सरल भाषा में स्पष्ट चित्रों के साथ प्रस्तुत
अनुदेशक गण द्वारा उत्तरों का स्पष्टीकरण ◉ प्रशिक्षुओं के कौशल में वृद्धीकारक
उत्तरों का स्पष्टीकरण सहित सॉल्वड पेपर ◉ सरल भाषा मे लिखित

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Sign bit extension:

- Sign
- 1's
- 2's

1's complement of +6

1 1 1 0 . (-6)

1 0 0 0 1 (-6)

1	0 0 0	1 1 0

Magnitude

1 1 1 1 1 0 0 1

(-6)

Eg. sign Representation in 2's complement form. FFF3. then decimal value.

1 1 1 1	1 1 1 1	1 1 1 1	0 0 1 1

We can discard these. (++)

Overflow Condition

- Unsigned
  - Carry Always Indicate Overflow.
- Signed
  - If Sign bit changed.
    - If two positive Added & Result is -ive.
    - If two -ive Added & Result is +ive.

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Binary Codes

→ self Complementary Nos
0101 - 2
1011 - 7 } 9 ✓

- ASCII
- EBCDIC
  - Alphanumeric
- Numeric
  - BCD (Binary decimal)
    - Weighted
      - 8421
      - 3321 etc.
    - Un-weighted
      - → Gray (Unit distance)
      - → Excess [self Comp]

Uses in Counters ++

To Avoid clock delays b/w two bits ++

Imp. Note for weighted eg. $(863)_{10}$ to 2241

8 6 3

1110 [1010 0110]

Two Combo.

6 ≥ 4 so,

9's compl = 3.

for 9's Compliment,

[1001 0101] → Least value this.

Then take 1's Compliment

1010 ++

So, 6 → 0101 Ans

[take 1's complement of this]

Imp:

if No. ≥ 4 then take 9's Complement.

if No ≤ 4 then check Least value in Binary form.

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सरल भाषा मे लिखित
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BCD Add^{n} :

if sum ≥ 9 or get carry then Add 6.

127	1	1	
789	0001	0010	0011
912	0111	1000	1001
✓		1011	1100 (12)
(A)		0110	+0110 [6]
		0001	0010

Exp.
→ If get carry & get Invalid form then Add 6.

Excess-3 :

If Carry occur ⇒ Add +3
If Carry not occur ⇒ ~~3~~ -3.

$x+3$
$y+3$
$z+6-3$.

7 → 07
3 → 03
10

✓ Gray Code:

Reflexive, Unit distance, Cycle [(0,15)(1,14)], Non-weighted.

Binary to Gray : 1000 → 1100.

Gray to Binary : 1010 → 1100.

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